STIVEN CATALYST

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MANAGEMENT REVIEW

Management without theatre.



STRATEGY

AFTER ACTION REVIEW: WHAT DID WE LEARN FROM THIS

An opponent that almost always wins, a review that is not a critique, what 46 samples of debriefs show, four questions before and after, why to review successes too, and a sheet for your own review.

25% better performance, on average, for teams and individuals who debriefed than for control groups, across 46 samples.

TANNENBAUM & CERASOLI, 2013

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Most teams finish a job and go straight on to the next one. This issue is about the short meeting in between: what was supposed to happen, what did happen, why, and what changes next time.

The US Army's Opposing Force almost always beats the brigades that come to train against it, and Marilyn Darling and colleagues traced that to its after-action reviews. The Army's 1993 guide defines the review and insists that it is not a critique. A meta-analysis of 46 samples put the average gain from debriefs at about 25%. Soldiers who reviewed their successes as well as their failures improved faster. And a lesson, in the Opposing Force's view, counts as learned only once it has been applied.

Stiven Janaqi EDITOR · STIVEN CATALYST

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HOW TO READ THIS ISSUE

Figure
Every figure has its source and year at the foot of its page.

Our reading
Where the editors interpret rather than the research, it says so.

Practice
The steps and the card are proposals to try, not research results.

COVER STORY

Not a critique

In the California desert, the US Army keeps a standing Opposing Force of 2,500 soldiers. Every month a fresh brigade of more than 4,000 takes it on, with more resources and better data, and it knows the Opposing Force's methods from earlier campaigns. Yet, wrote Marilyn Darling, Charles Parry and Joseph Moore in 2005, the Opposing Force almost always wins.

The authors trace the difference to the after-action review. The Army's 1993 guide defines it as a professional discussion of an event, focused on standards, in which soldiers discover for themselves what happened, why, and how to sustain strengths and improve on weaknesses.

“An AAR is not a critique.”

01
Everyone speaks
no one, whatever their rank, has all the information or the answers

02
Open questions
“what happened when...?” rather than “why didn't you...?”

03
No grades
there are always strengths to sustain and weaknesses to improve

OUR READING

A critique tells people what they did wrong. A review lets them find it themselves, which is why they remember it.

Sources: Marilyn Darling, Charles Parry & Joseph Moore, Harvard Business Review 83(7), 2005; Headquarters, Department of the Army, 1993

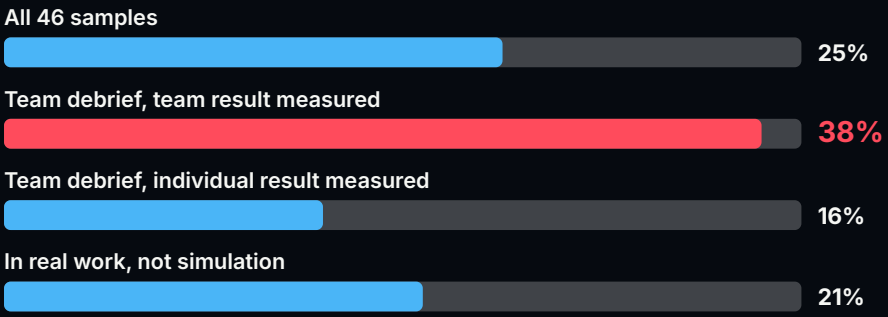
The Opposing Force's size and record are as the 2005 article gives them. The definition and the three cards follow TC 25-20 (1993), in our words; the Army issued a newer guide in 2013 (page 5).

THE NUMBERS

About a quarter better

In 2013 Scott Tannenbaum and Christopher Cerasoli pooled 46 samples with 2,136 participants: teams and individuals, simulations and real work, medicine and other fields. On average, those who debriefed performed about 25% better than those who did not.

AVERAGE GAIN OVER THE CONTROL GROUP, BY KIND OF DEBRIEF, 2013



The debriefs studied lasted about 18 minutes on average, and longer sessions were not more effective. In 2021 Nathanael Keiser and Winfred Arthur, across 61 studies, found an even larger average effect.

OUR READING

A review does not need to be long. It needs to measure the same thing it set out to improve.

Sources: Scott I. Tannenbaum & Christopher P. Cerasoli, Human Factors 55(1), 2013; Nathanael L. Keiser & Winfred Arthur Jr., Journal of Applied Psychology 106(7), 2021 (via Abstract; QIC-WD research summary, 2021)

The percentages are the authors' conversion of the effect size (d = 0.67 overall; 0.79 in Keiser & Arthur). Only 6 samples came from real work. Many studies were not randomised, so the authors urge care about cause.

THE MODEL

Before and after the action

The Army's 2013 guide gives every review the same four parts, in training and in operations. The answers to the last one name who is responsible for each change.

1. **What was supposed to happen?**
   The intent, the objectives and the standard, as the plan stated them.
2. **What happened?**
   From as many viewpoints as possible, until everyone shares one picture.
3. **What was right or wrong with it?**
   Strong and weak points, measured against the intent and the standard.
4. **What will we do differently next time?**
   The team finds its own solutions and names who makes each change.

Darling, Parry and Moore start the cycle earlier, with a before-action review of four questions: What are our intended results and metrics? What challenges do we anticipate? What have we or others learned from similar projects? What will enable us to succeed this time?

OUR READING

The review before the action writes the first answer of the review after it: what was supposed to happen.

Sources: U.S. Army Combined Arms Center, Training Management Directorate, 2013; Headquarters, Department of the Army, 1993; Marilyn Darling, Charles Parry & Joseph Moore, Harvard Business Review 83(7), 2005

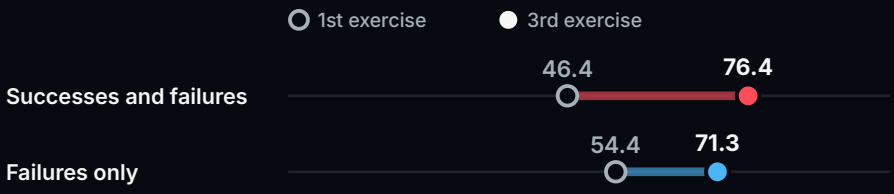
The four parts follow the Army's 2013 guide, in our words; the 1993 circular follows a similar sequence. The before-action questions are quoted from Darling et al. The reading is the editors'.

WHAT THE RESEARCH SAYS

Review the successes too

Shmuel Ellis and Inbar Davidi followed 98 soldiers of an Israeli elite unit through a navigation course. After every exercise a commander reviewed each soldier one to one, for about 14 minutes: in one company only what had gone wrong, in the other failures and successes.

MEAN NAVIGATION SCORE, THREE EXERCISES IN THE SECOND WEEK, 2005



The company that also reviewed its successes improved significantly faster, though the exercises got harder each day. At first the soldiers explained failures in more detail than successes; where both were reviewed, the gap closed. In a 2021 meta-analysis, team reviews of team results worked best when the team led them.

OUR READING

A success that nobody examines is a success nobody can repeat on purpose.

Sources: Shmuel Ellis & Inbar Davidi, Journal of Applied Psychology 90(5), 2005; Nathanael L. Keiser & Winfred Arthur Jr., Journal of Applied Psychology 106(7), 2021 (via Abstract; QIC-WD research summary, 2021)

A quasi-experiment: whole companies were assigned, not soldiers. The score weights points reached, pace and map use; only the second week was compared. Keiser & Arthur: abstract.

HOW IT IS MEASURED

A lesson counts when it is applied

After-action reviews became a popular business tool after Shell began experimenting with them in 1998, Darling and colleagues write, but in most companies they fall short: lessons are drawn, and rarely learned. The Opposing Force counts a lesson as learned only once it has been applied and validated.

01	**Changes done on time**	Of the changes agreed in the fourth question, how many were made by the date set.
02	**Repeat findings**	Issues that come back in a later review: a lesson drawn, but not yet learned.
03	**The result you reviewed**	A team review is judged by the team's result, not by one person's.

HYPOTHETICAL EXAMPLE, A MONTH OF REVIEWS IN A WAREHOUSE

Reviews	8 planned, 6 held
Changes	14 agreed, 8 done by the date
Repeats	3 issues already raised in an earlier review

The last line shows lessons drawn but not learned. The numbers are invented.

Sources: Marilyn Darling, Charles Parry & Joseph Moore, Harvard Business Review 83(7), 2005; Scott I. Tannenbaum & Christopher P. Cerasoli, Human Factors 55(1), 2013

Shell and the Opposing Force's rule are from Darling et al. (2005); judging a team review by the team's result follows Tannenbaum & Cerasoli. The three counts and the example are the editors'.

More in the essay When improvement fades

TOOL OF THE ISSUE

The after-action review sheet

Fill it in soon after the event, with the people who took part. Write facts, not blame, and leave no line in the last box without a person and a date.

01 Event and who took part one event or phase, not a whole month

02 What was supposed to happen the goal, the plan and the standard, as they stood before

03 What actually happened the facts, from more than one point of view

04 What went well, and why what we keep, so we can repeat it on purpose

05 What did not, and why causes, not culprits; ask why until you reach something you can change

06 What we do differently next time each change with an owner and a date; check it at the next review

Sources: U.S. Army Combined Arms Center, Training Management Directorate, 2013; Shmuel Ellis & Inbar Davidi, Journal of Applied Psychology 90(5), 2005; Marilyn Darling, Charles Parry & Joseph Moore, Harvard Business Review 83(7), 2005

A practice proposed by the editors, after the Army's four questions, the review of successes and failures in Ellis & Davidi, and Darling et al.

Open the tool: 5 Whys stivencatalyst.com/tools/five-whys/

SOURCES AND METHOD

Every figure has a source.

The figures in this issue come from the sources below. The year shows how recent each one is.

Learning in the Thick of It
Marilyn Darling, Charles Parry & Joseph Moore, Harvard Business Review 83(7), 2005
https://hbr.org/2005/07/learning-in-the-thick-of-it

TC 25-20: A Leader's Guide to After-Action Reviews
Headquarters, Department of the Army, 1993

The Leader's Guide to After-Action Reviews (AAR)
U.S. Army Combined Arms Center, Training Management Directorate, 2013

Do Team and Individual Debriefs Enhance Performance? A Meta-Analysis
Scott I. Tannenbaum & Christopher P. Cerasoli, Human Factors 55(1), 2013
https://doi.org/10.1177/0018720812448394

A meta-analysis of the effectiveness of the after-action review (or debrief) and factors that influence its effectiveness
Nathanael L. Keiser & Winfred Arthur Jr., Journal of Applied Psychology 106(7), 2021 · via Abstract; QIC-WD research summary, 2021
https://doi.org/10.1037/apl0000821

After-Event Reviews: Drawing Lessons From Successful and Failed Experience
Shmuel Ellis & Inbar Davidi, Journal of Applied Psychology 90(5), 2005
https://doi.org/10.1037/0021-9010.90.5.857

EDITORIAL METHOD

Each figure was checked for its year, its publisher and what exactly it measures. Where the publisher's page could not be opened, the figure was checked against independent summaries and is marked “via”. The editors' interpretation is marked “Our reading”. Figures that could not be confirmed are not in the issue.



MANAGEMENT REVIEW

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Every issue, one management question, checked against the best research.

ALL ISSUES

stivencatalyst.com/magazine/management-review.html

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